

Mohammad Kazzazi

✉ smohammadok@gmail.com ☎ +30 697 672 22 37 🔗 LinkedIn 🐙 GitHub 🎓 Google Scholar

Education

École Polytechnique Fédérale de Lausanne (EPFL)

M.Sc. in Electrical Engineering

Sept 2026 – Present

Lausanne, Switzerland

Sharif University of Technology – GPA: 18.53/20 (3.95/4)

B.Sc. in Electrical Engineering (Digital Systems)

Sept 2021 – 2025

Tehran, Iran

Research Interests

- Efficient Deep Learning & On-Device AI
- AI / FPGA Accelerators for Neural Networks
- Computer Vision & Vision-Language Models
- GPU Computing & Parallel Architectures
- Deep Learning Optimization
- AI Systems & Hardware Acceleration

Research Experience

Research Assistant – Aalto University

Supervisor: Prof. S. Khajavi

Apr 2026 – Present

Espoo, Finland (remote)

- Developed a **knowledge-distillation framework** (KL + cross-entropy) that compresses a 7B Qwen2.5 vision-language teacher into 0.5B–3B students for **on-device deployment**.
- Applied it to contextual fire understanding on the DetectiumFire benchmark: the 0.5B student keeps **95.2%** of the teacher's burning-object accuracy with **14× fewer parameters** and reaches **99.5%** fire/non-fire accuracy after deployment.
- Deployed and profiled on a **Raspberry Pi 5** (4× Cortex-A76, 8 GB, CPU only): **1.9 GB peak memory**, ≈ 4 s end-to-end latency, 2 false alarms on 1,452 test images. First-author submission to **AAAI 2027 IAAI**.
- Currently investigating prompt optimization and reasoning in **Vision-Language-Action (VLA)** models.

Research Assistant – Raderon AI Laboratory

Supervisor: Dr. S. Sarabi

Aug 2024 – Jan 2025

British Columbia, Canada (remote)

- Conducted ML research on medical image analysis combining the Discrete Wavelet Transform with swarm-based optimizers; co-authored a paper published in Elsevier *Machine Learning with Applications*.

Research Assistant – Artificial Creature Laboratory (ACLab), Sharif University

Supervisor: Prof. S. Bagheri Shouraki

Jul 2024 – Sept 2024

Tehran, Iran

- Designed and built a **smart IoT socket** for the lab's custom AI platform, with a path to commercialization.
- Evaluated IoT platforms and network protocols to select the stack; implemented firmware on **Arduino** and **ESP32/ESP8266**.

Publications

1) Distilling Vision-Language Models for On-Device Fire Understanding

Sept 2026 – Aalto University, Finland | Submitted to AAAI 2027 Innovative Applications of AI (IAAI)

Authors: **Mohammad Kazzazi**, Zixuan Liu, S. Khajavi

[arXiv:2609.05782](#)

2) Minimal-Sensor TDOA/FDOA Fusion for Moving-Source IoT Localization: A Closed-Form, Two-Stage, Low-Complexity Estimator

Oct 2025 – Sharif University of Technology | Submitted to IEEE Transactions on Aerospace and Electronic Systems

Authors: **Mohammad Kazzazi**, Mohammad Morsali, R. Amiri

[arXiv:2510.04160](#)

3) Enhancing SCD Using Late Discrete Wavelet Transform and New Swarm-Based Optimizers

Nov 2024 – Raderon AI Lab, Canada | Published in Elsevier *Machine Learning with Applications*

Authors: R. Mousa, M. Morsali, S. Chamani, **M. Kazzazi**, P. Hatami, S. Sarabi

DOI: [10.1016/j.mlwa.2025.100811](#)

Selected Projects

more on [GitHub](#)

Football Object Detection and Multi-Object Tracking

[code](#)

- Fine-tuned **YOLOv8** on the SportsMOT football split (720p, 25 fps): **mAP@50 0.947**, mAP@50–95 0.779; 1.4 ms inference per frame on a P100 GPU. Added **ByteTrack**, **DeepSORT** (MOTA / IDF1) and **CSRT** tracking.

Membership Inference Attacks Against Diffusion Models

 [code](#)

- Re-implemented **SecMI** (ICML 2023) on CIFAR-100 DDPMs and proposed a hyperparameter-free multi-timestep variant: **AUC 0.971** $\rightarrow$ **0.984**, TPR@1%FPR 0.519 $\rightarrow$ 0.642.

Machine Unlearning and Private Training

 [code](#)

- **SISA** sharded training with ResNet-18 on CIFAR-10 (92% accuracy); unlearned 500 samples and verified removal via backdoor attack-success rate and membership inference.

Hardware & Parallel Computing

- [CORDIC Algorithm on Zynq FPGA](#) (Verilog, Vivado)
- [FFT on Zynq FPGA with VGA output](#) (Verilog)
- [Multi-cycle MIPS Processor](#) (Verilog)
- [Parallel Merge Sort on GPU](#) (CUDA)
- [Heat-Transfer Simulation with CUDA Texture Memory](#)
- [Digital Telecommunication System Simulation](#)

Technical Skills

Programming: Python, C/C++, Verilog, CUDA, Java, MATLAB, Assembly

Hardware & EDA: Vivado, Xilinx ISE, ModelSim, LTspice, Zynq FPGA, Raspberry Pi, Arduino, ESP32/ESP8266

ML / CV: PyTorch, Hugging Face Transformers, OpenCV, Ultralytics YOLO, scikit-learn, NumPy, pandas, knowledge distillation, on-device inference

Tools: Git, Linux, MySQL, $\LaTeX$

Languages: Persian (native), English (TOEFL iBT 105), Spanish (A1)

Relevant Coursework

EPFL – Fall 2026 (in progress)

- Machine Learning
- Mathematics of Data
- Convex Optimization
- Fundamentals of Inference and Learning

Sharif University of Technology – graduate courses (grades out of 20)

- VLSI – 19.1
- Parallel Computing – 18.4
- Convex Optimization II – 19.7
- Deep Learning – 17.2
- Deep Generative Models – 17.2
- Security in IoT – 17.8

Sharif University of Technology – undergraduate courses (grades out of 20)

- ASIC/FPGA System Design – 17.2
- Data Structures and Algorithms – 19.5
- Linear Algebra / Mathematical Methods – 20
- Communication Systems – 19
- Introduction to Machine Learning – 17.9

Teaching Experience

Laboratory Teaching Assistant – Sharif University of Technology

Fall 2023

- Logic Circuits and Digital Systems Lab.

Teaching Assistant – Sharif University of Technology

2022 – 2025

- Logic Circuits and Digital Systems (4 terms, 2023–2024), Engineering Probability and Statistics (Fall 2025), Mathematical Methods in Engineering (Spring 2024), Communication Systems (Spring 2024), Object-Oriented Programming (Spring 2023), Basic Programming (Fall 2022).

Awards & Academic Distinctions

- Ranked in the **top 0.1%** of approximately 180,000 participants in Iran's National University Entrance Exam; admitted to Sharif University of Technology with a fully funded scholarship.
- **Top 15%** of the B.Sc. Electrical Engineering class and **top 3** in the Digital Systems specialization.

References

Prof. R. Amiri

Associate Professor, Electrical Engineering
Sharif University of Technology
amiri@sharif.edu

Prof. S. Bagheri Shouraki

Full Professor, Electrical Engineering
Sharif University of Technology
bagheri-s@sharif.edu

Prof. S. Khajavi Haghighat

Assistant Professor, Industrial Engineering and Management
Aalto University
siavash.khajavi@aalto.fi

Dr. S. Sarabi

Director, Raderon AI Lab
s.sarabi@raderonlab.ca